

Recall the General Balance Equation:

$$\text{Accumulation} = \text{In-Out} + \text{Generation} - \text{Consumption}$$

When we're dealing with mass and energy,

$$\text{Accumulation} = \text{In} - \text{Out}$$

When we're dealing with moles,

When we're dealing with moles,
Generation and consumption terms are valid

Steady state system:

Accumulation $\neq \emptyset$

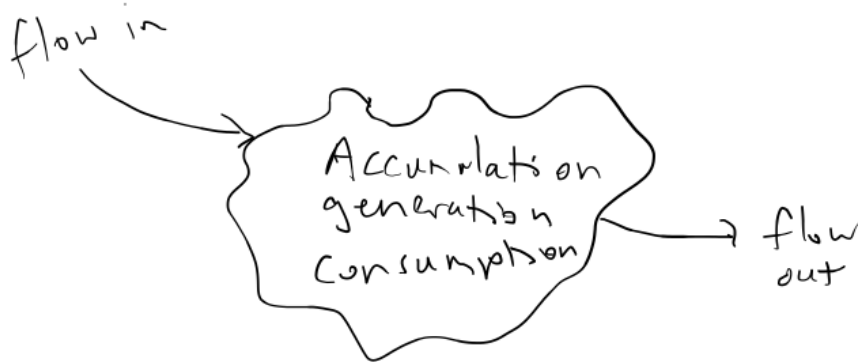
Batch system:

Batch system:

In $\sum_1$ out terms = $\emptyset$

The General Balance Equation in CRE:

- Written over a volume
- Written for chemical species, e.g.) CO(g) , or species A



- In CRE, generation and consumption terms are lumped into one "net generation term. G_j "
 - Consider species j . If species j is net consumed:
 G_j should be negative
 - If species j is net generated:
 G_j should be positive
 - KEY POINT: This is an accounting issue. Use logic to ensure proper accounting of the various species.

Let's derive the Mole Balance for species j :

Accumulation = In - Out + G_j

$$\frac{dN_j}{dt} = F_{j0} - F_j + G_j \quad \text{Units are moles/time}$$

How we treat the G_j term depends on what is happening with the system variables.

- What system variables are relevant for CRE? concentration, temperature, catalytic activity
- If all the system variables are uniform

- The system is said to be well mixed

- $G_j = r_j V$

- r_j has units of $\frac{\text{moles}}{\text{volume} \cdot \text{time}}$

- Else: $G_j = \int_V r_j dV \rightarrow$ integrating over all of the rates in the volume.
 $\uparrow$ r_j is a function of volume

Examples of systems where the variables are non-uniform:

non perfect mixing in a reactor
non ideal behavior
Lake Hartwell